# Separated voices and speaker timelines

Simulated test conversation: real Hindi speech, two speakers, heavy overlap and 5 dB market noise.

## M1 Separation / Vismay

Conv-TasNet output clips, 28.0–35.5 s




**+9.6 dB SI-SDR improvement on 245 test overlap regions**

## M2 Diarization / Darshan

Actual timeline.json output (excerpt)

| Start (s) | End (s) | Speaker |
|---|---|---|
| 26.85 | 31.75 | Speaker_A |
| 31.75 | 33.20 | Speaker_B |
| 33.20 | 40.20 | Speaker_A |

**20.3% DER**

Spectral clustering, 72 test conversations
GMM: 20.3% / pyannote 3.1: 18.4%

# What the system actually transcribes

M3 / Abhinav  IndicConformer speech recognition, Hinglish text and subtitle files

SAVED HINGLISH TRANSCRIPT

**00:27–00:32 Speaker_A**

sir mere koi galti hui mere haath se wo mera aai D card gum ho gaya hai sir

**00:32–00:33 Speaker_B**

haan aur baki sun

**00:51–00:54 Speaker_A**

to ab kab tak kitne baje tak office khula hai sir apna

Devanagari + Hinglish + SRT + JSON

## 21.0% word error

IndicWav2Vec: 38.0%
400 Vaani clips, single-speaker ASR

FULL PIPELINE / 72 CONVERSATIONS

| Order | cpWER |
|---|---|
| A: separate first | 55.8% |
| B: diarize first | 49.5% |

B wins in 50 of 72 conversations.
cpWER measures words and speaker attribution. Lower is better.

# The report output, and what remains

M4 / Shrivaths  Airavata 7B, quantised to 4 bits and run locally

ACTUAL REPORT EXCERPTS

Key point

“Speaker A lost his ID card”

Line 13 / Speaker_A / 00:51–00:54

“The office is open until what time?”

Produces a summary, keywords and translated dialogue.
Speakers and times pass through unchanged.

## Measured limits

Transcript repair adds 0.9 cpWER points
(paired mean across 63 conversations).
No accepted action items in 252 reports.

## Next steps

Improve Hindi speaker embeddings.
Evaluate a larger LLM for action items.
Retrain separation on Indian voices and noise.

These are planned extensions, not measured results.